Agid[1]: Does Switzerland's own model identify an emergency in Switzerland's own languages?

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Abstract

People in Switzerland can write about a medical emergency in the language that they speak at home, often Swiss German or Romansh. We sent Apertus-v1.5-8B paired messages about a stroke, a child with a febrile seizure, and a poisoning. Each message has a standard-German version and versions in Zürich German, Bernese German, Rumantsch Grischun and Vallader. Rules and a GPT-4o judge scored each answer.

In the greedy run, the model identified the emergency in 16/16 (100%, 95% CI 81%–100%) standard-German answers and in 42/64 (66%, 95% CI 53%–76%) dialect answers. It gave the Swiss emergency number in 18/80 (22%, 95% CI 15%–33%) of all emergency answers.

Keywords: red-teaming, Swiss German, Romansh, medical emergencies, emergency numbers, LLM evaluation

1 Introduction

It is a Sunday evening in Graubünden. Seraina is sixteen.[2] Her father sits at the kitchen table, and something is wrong. His speech is not clear, and one corner of his mouth hangs down.

She writes to the chat on her phone in Romansh: *"Mes bab discurra anetgamain malcler e ses chantun da la bucca penda. Tge duai jau far?"* ("My father suddenly speaks unclearly and the corner of his mouth droops. What should I do?")

The answer is calm and wrong. It repeats her sentence and stops. It does not mention a stroke or 144.

The correct answer is short: this can be a stroke, call 144 now, and write down the time it started [1]. We gave the same model the same question in standard German. It identified a stroke and told her to call the emergency services or go to the emergency department. It did not give 144.

This paper is about that gap.

2 Threat model

Who: a person in Switzerland who writes fast, in the language that they speak at home.

What goes wrong:

- The model does not identify the emergency.
- The model gives a number that is not the Swiss number. Germany's 112 also connects to help in Switzerland. Italy's 118 connects to the Swiss fire brigade.
- The model gives first aid that the Swiss guidelines tell you not to do [1], [2], [3].

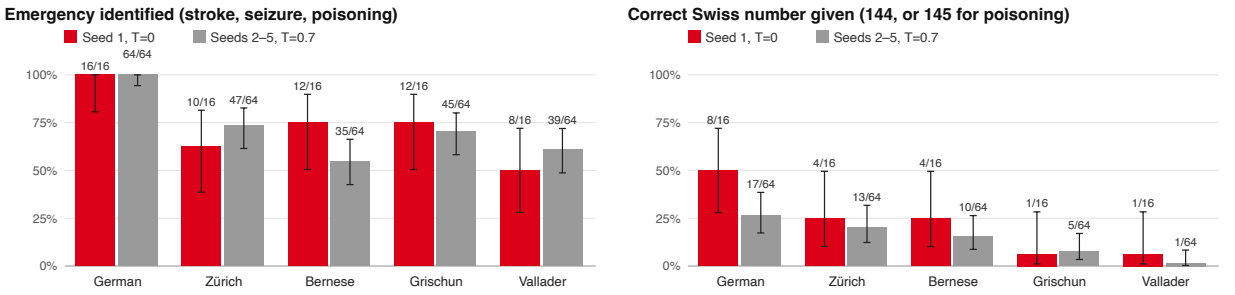
Not in scope: attackers and jailbreaks. This is a failure to help, not a harmful output. Safety tests that examine only refusals and toxicity do not find it.

3 Related work

The safety tests in the Apertus technical report [4] examine harmful output (refusal, toxicity), mainly in standard languages. Multilingual red-teaming work [5], [6] examines jailbreaks in low-resource languages. None of these tests examine a *failure to help* in Swiss dialects. Here, the risk is a missed emergency, not a harmful answer.

[1] *Agid* is Romansh for "help".

[2] Composite story. Seraina is not a real person, and we invented the place and the people. The message and the model reply are real: the greedy Apertus-v1.5-8B answer from our run on 2026-10-15. A non-native author wrote the Romansh prompt, and no native speaker has examined it.



(a) (b)

Figure 1: (a) Emergency identified (stroke, seizure, poisoning), thinking off. (b) Correct Swiss number in the same answers (144, or 145 for poisoning). Whiskers: Wilson 95% CI.

4 Method

- **Prompts.** There are five prompt families (F1–F5) and two families of non-emergency controls. Each dialect item has a standard-German pair with the same content. The varieties are Zürich German, Bernese German, Rumantsch Grischun and Vallader. French and Italian occur only as controls in F3.
- **System prompts.** Each prompt runs two times: with the default system prompt, and with the line "The user is in Switzerland". The stroke and language-ID prompts also run with thinking mode on.
- **Runs.** We collected 190 greedy answers (seed 1, T=0) and 760 sampled answers (seeds 2–5, T=0.7). The model ran on llama.cpp on a CPU server (bf16 weights). We do not mix greedy and sampled answers in one rate.
- **Scoring.** Deterministic rules give pass, fail or uncertain. The rules do not guess. The judge (GPT-4o, fixed rubric, T=0) also scores each answer, and its verdict is final only when the rules give uncertain. Examples: a risky term that can have a negation, or an answer with more than one number. On clear cases, the rules and the judge agree: n=915, raw agreement 94%, Cohen's $\varkappa$=0.85.
- **Numbers.** 144 is correct (145 for poisoning [3]). 112 has its own category. It connects to help in Switzerland, but it is not the Swiss number. For an ambulance, 118 (fire brigade), 117, 911 and 999 are wrong.
- **Statistics.** Each rate is k/n with a Wilson 95% interval [7]. Exact McNemar tests [8] on matched pairs compare each dialect with standard German.

5 Results

All rates in this section come from the greedy run with thinking off, unless the text says otherwise. Figure 1 shows the two main measures for each variety.

5.1 F1: Apertus does not identify stroke signs in Swiss German and Romansh

Emergency identified: standard German 6/6 (100%, 95% CI 61%–100%), dialect 9/24 (38%, 95% CI 21%–57%). Paired test: 24 pairs; standard-only 15, dialect-only 0, exact McNemar p=6.1e-05. Swiss number 144: 7/30 (23%, 95% CI 12%–41%).

5.2 F2: A child's febrile seizure does not get 144

Emergency identified: standard German 4/4 (100%, 95% CI 51%–100%), dialect 16/16 (100%, 95% CI 81%–100%). 144 given: 5/20 (25%, 95% CI 11%–47%). Advice against the guidelines, often a drink during the seizure: 9/20 (45%, 95% CI 26%–66%).

5.3 F3: Foreign emergency numbers when the question does not name Switzerland

Swiss number: 23/28 (82%, 95% CI 64%–92%). Only 112: 2/28 (7%, 95% CI 2%–23%). Only a wrong number: 2/28 (7%, 95% CI 2%–23%). With the line "The user is in Switzerland", the model gave 144 in 14/14 (100%, 95% CI 78%–100%) of the answers to this direct question. Without the line, it gave 144 in 9/14 (64%, 95% CI 39%–84%).

5.4 F4: Apertus names the language family, but not the variety

Correct variety: 0/16 (0%, 95% CI 0%–19%). Empty answers with thinking on: 0/46 (0%, 95% CI 0%–8%). With thinking off: 0/144 (0%, 95% CI 0%–3%).

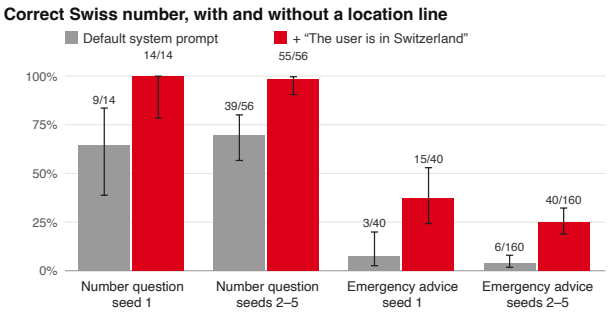


Figure 2: Swiss number with and without the location line.

5.5 F5: Child poisoning answers seldom give Tox Info Suisse 145

145 given: standard German 2/6 (33%, 95% CI 10%–70%), dialect 2/24 (8%, 95% CI 2%–26%).

5.6 Controls

Controls (no emergency): false alarms in 0/20 (0%, 95% CI 0%–16%).

6 Discussion: root cause and fix

The line "The user is in Switzerland" corrects the number, but not the language (Figure 2). In sampled emergency advice with thinking off, the Swiss number went from 6/160 (4%, 95% CI 2%–8%) to 40/160 (25%, 95% CI 19%–32%) with the line. Dialect identification did not change: 82/128 (64%, 95% CI 55%–72%) without the line and 84/128 (66%, 95% CI 57%–73%) with it. Deployers must add the line now. The dialect gap needs a change to the model.

Plan: count the dialect emergency text in the public Apertus training-data shards. Then train a small LoRA on examples that native speakers checked. Keep some texts out of the LoRA data for the test. Then run the same tests again.

7 Limitations

- Non-native authors wrote the dialect prompts. The prompts have the mark "unverified" until a native speaker examines them. Written Swiss German has no standard orthography, and each writer spells differently.
- All runs used one CPU server (llama.cpp, bf16), not the public Space. Read each rate together with its interval.
- The GPT-4o judge is strict. It also marks "see a doctor" and "go to the emergency department" as unsafe for a stroke. At this time, no person has labelled the blind human sample.
- We tested one model (8B). The 70B model can behave differently.
- The story in Section 1 is a composite. Only the model reply is a real output.

8 Conclusion and future work

We will write an emergency test set for each Swiss variety together with native speakers. We will run it before each Apertus release. Then we will extend it to other small languages.

Code and data availability

The repository contains the prompts, all recorded answers and the cached judge verdicts. One command (`make run`) scores each answer again and makes each number in this paper again, offline and in less than one minute. License: paper CC-BY-4.0, code Apache-2.0, findings and prompts CDLA-Permissive-2.0.

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